



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

29-08-2021

JEE(Main + Advanced) : Enthusiast Course (PHASE : S, I(A), I, LIVE-I & I(A))

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	D	A	A	D	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	C	D	B	A	C	D	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	14.50	200.00	7.00	2.00	0.00	5.00	1.60	1.00	5.00	6.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	D	C	C	B	D	D	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	D	B	A	D	D	B	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2.00	33.33	-1.00	6.00	3.00	3.00	8.00	3.00	5.00	4.00

PART-3 : MATHEMATICS

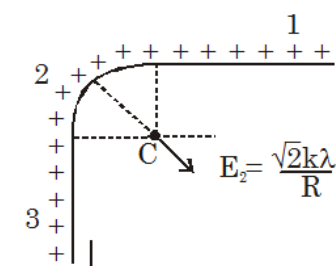
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	A	C	C	A	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	B	A	D	D	C	A	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3.00	47.08	4.33 or 4.34	7.00	4.00	2.41 or 2.42	3.00	9.00	4.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)



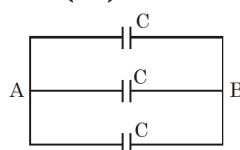
$$\vec{E}_2 = \frac{k\lambda}{R} \hat{i} - \frac{k\lambda}{R} \hat{j}$$

$$\vec{E}_1 = -\frac{k\lambda}{R} \hat{i} - \frac{k\lambda}{R} \hat{j}$$

$$\vec{E}_3 = \frac{k\lambda}{R} \hat{i} + \frac{k\lambda}{R} \hat{j}$$

$$\vec{E}_{\text{net}} = \frac{k\lambda}{R} \hat{i} - \frac{k\lambda}{R} \hat{j}$$

2. Ans (C)



$$C = \frac{A\epsilon_0}{2d}$$

$$C_{AB} = 3C$$

5. Ans (A)

$$mg = B = v\rho g$$

$$xL^2 = \frac{m}{\rho_L} = \frac{L^3 \rho_s}{\rho_L}$$

$$x = \frac{\rho_s}{\rho_L} L$$

6. Ans (A)

$$\text{Let at } P(x, y) \quad B = 0$$

$$\frac{2ki_1}{y} = \frac{2ki_2}{x}$$

$$5y = 10x$$

$$y = 2x$$

8. **Ans (B)**

$$\Rightarrow B_{\text{net}} = \frac{\mu_0 i}{4\pi r} \theta$$

9. **Ans (B)**

$$\Rightarrow B_{\text{net}} = 0 + \frac{ki}{r} + 0 + \frac{ki}{r} = \frac{2ki}{r}$$

10. **Ans (A)**

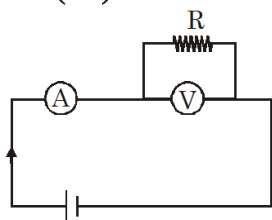
$$i = \frac{20}{3+2} = 4\text{A}$$

$$i_A = i_B = 2\text{A}$$

$$V_A + 2 \times 1 - 3 \times 2 = V_B$$

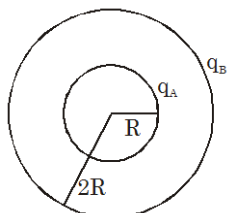
$$V_A - V_B = 4\text{V}$$

11. **Ans (D)**



when a resistance is joined in parallel with voltmeter, effective resistance of circuit will decrease and current through ammeter increases so, A increases. The sum of A & V is constant so, V decreases.

13. **Ans (C)**



$$\frac{kq_A}{R} + \frac{kq_B}{2R} = 2V$$

$$\frac{kq_A}{2R} + \frac{kq_B}{2R} = \frac{3}{2}V$$

$$\Rightarrow \frac{q_A}{q_B} = \frac{1}{2}$$

After Earthing, charge on A remains same. Let charge on B be q'_B

$$\Rightarrow \frac{kq_A}{2R} + \frac{kq'_B}{2R} = 0$$

$$= q_A = -q'_B$$

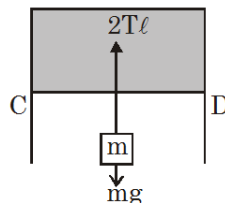
$$= \frac{q_A}{q_B} = -1$$

$$= V_A = \frac{kq_A}{R} + \frac{kq'_B}{2R} = \frac{V}{2} \text{ \& } V_B = 0$$

14. **Ans (D)**

$$\frac{V_1}{V_2} = \frac{C_2}{C_1} = \frac{K_2 \frac{A\epsilon_0}{2d}}{K_1 \frac{A\epsilon_0}{d}} = \frac{3}{2}$$

15. **Ans (B)**



For equilibrium, $2Tl = mg$

$$m = \frac{2Tl}{g}$$

16. **Ans (A)**

$$v = \sqrt{2g(H-h)}$$

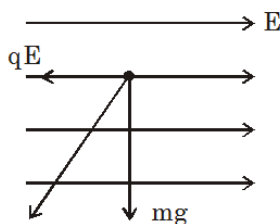
18. **Ans (D)**

Magnetic field on axis of circular coil is

$$\frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}} = \frac{\mu_0 i_1 R^2}{2(R^2 + x^2)^{3/2}} = \frac{\mu_0 i_2 R^2}{2(R^2 + x^2)^{3/2}}$$

$$x = \sqrt{7}R$$

19. **Ans (D)**



SECTION-II

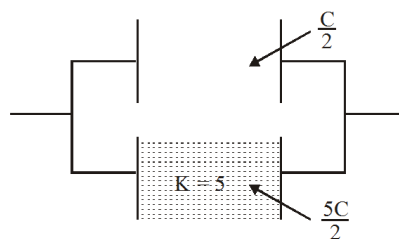
1. **Ans (14.50)**

$$|E| = \sqrt{29}$$

$$U = \frac{1}{2} \epsilon_0 E^2 \times l^3 = \frac{29\epsilon_0}{2}$$

2. **Ans (200.00)**

Now



$$\text{if } C = \frac{A\epsilon_0}{d}$$

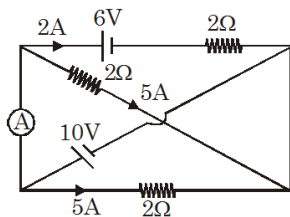
$$C_{\text{eq}} = 3C$$

$$\Delta C = 3C - C$$

$$\% \Delta C = 200\%$$

3. **Ans (7.00)**

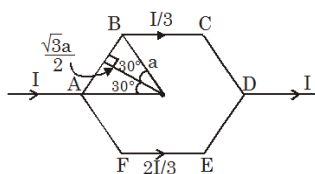
From diagram



Current through ammeter

$$= (2 + 5) = 7A$$

4. **Ans (2.00)**



$$B_{\text{due to ABCD}} = \frac{\mu_0 I}{4\pi a\sqrt{3}} (\sin 30^\circ + \sin 30^\circ) \times 3$$

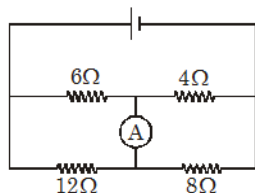
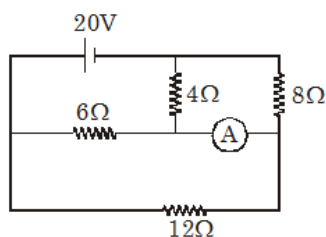
$$= \frac{2\mu_0 I}{a\sqrt{3}} \otimes$$

$$B_{\text{due to AFED}} = \frac{\mu_0 I}{4\pi a\sqrt{3}} (\sin 30^\circ + \sin 30^\circ) \times 3$$

$$= \frac{4\mu_0 I}{a\sqrt{3}} \odot$$

$$B_{\text{net}} = |B_1 - B_2| = \frac{2\mu_0 I}{a\sqrt{3}} = 2 \cdot \frac{\mu_0 I}{4\pi a\sqrt{3}}$$

5. **Ans (0.00)**



$$\frac{6}{12} = \frac{4}{8}$$

So, it will act as wheatstone bridge, current through ammeter $\rightarrow$ zero

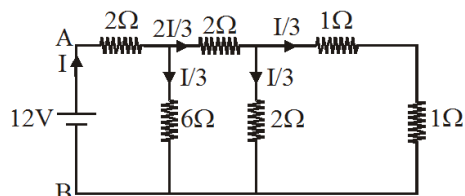
6. **Ans (5.00)**

$$\rho v_b g = 100 B = \rho_w v_b g = 100 - 80 = 20$$

7. **Ans (1.60)**

$$t = R_{\text{eq}} C_{\text{eq}} \\ = 2 \times 10^6 \times 0.8 \times 10^{-6} \\ = 1.6 \text{ sec}$$

8. **Ans (1.00)**

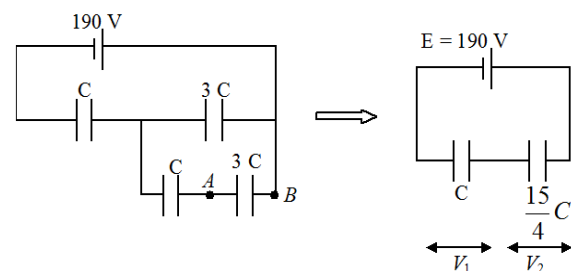


The equivalent resistance between A and B = 4Ω .

$$\text{Current } I = \frac{12V}{4\Omega} = 3A$$

The current divides into different branches as shown in the figure. It is clear that the ammeter will read $\frac{I}{3} = \frac{3A}{3} = 1A$. So the correct choice is (1)

9. **Ans (5.00)**



$$\frac{V_1}{V_2} = \frac{\frac{15}{4}C}{C} = \frac{15}{4},$$

$$V_1 + V_2 = 190$$

$$V_2 = 40 \text{ V}$$

This V_2 is distributed in C and $3C$ in the ratio of 3 : 1.

$$\therefore V_{AB} = 10 \text{ V}$$

10. **Ans (6.00)**

The resistors of 4Ω and 8Ω are connected in parallel.

$\therefore$ Potential drop across 4Ω resistor = Potential drop across 8Ω resistor

Let current in 4Ω resistor be i

$$\therefore 4i = 8 \times 2$$

$$\text{or } i = 4A$$

$$\therefore \text{current through } 2\Omega \text{ resistor} = 4 + 2 = 6A$$

PART-2 : CHEMISTRY

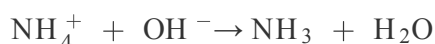
SECTION-I

1. **Ans (B)**

$$5 = 4.7 + \log \frac{[\text{NH}_4^+]}{0.25}$$

$$[\text{NH}_4^+] = 0.5 \text{ M}$$

m moles of KOH added = 50



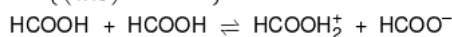
$$\begin{array}{cccc} 100 & 50 & 50 & \text{excess} \\ 50 & 0 & 100 & \text{excess} \end{array}$$

$$\text{pOH} = \text{pK}_b + \log \frac{50}{100} = 4.4$$

$$\text{or pH} = 14 - 4.4 = 9.6$$

2. **Ans (D)**

$$[\text{HCOOH}] = \frac{1 \text{ mol}}{\left\{ \left(\frac{46}{1.15} \right) \times 10^{-3} \ell \right\}} = 25 \text{ M}$$



$$\begin{array}{ccccc} t = 0 & 25 \text{ M} & 25 \text{ M} & 0 & 0 \\ \text{At equilib.} & 25 - 10^{-3} & 10^{-3} & 10^{-3} & 10^{-3} \end{array}$$

$$\alpha (\%) = \frac{10^{-3}}{25} \times 100 = 0.004\%$$

3. **Ans (D)**

$$\text{For ideal gas } \Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$\text{So, } \Delta S = 1C_v \ln \frac{2T}{T} + 1R \ln \frac{V/2}{V}$$

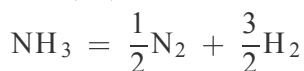
$$\Delta S = C_v \ln 2 - R \ln 2$$

$$\Delta S = (C_v - R) \ln 2$$

5. **Ans (C)**

$$\Delta H = q_p$$

6. **Ans (B)**

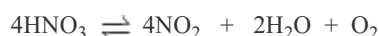


V ↑

Eq.

→ (α ↑)

7. **Ans (D)**



$$\begin{array}{cccc} \text{equ. 2atm} & 4\text{P} & 2\text{P} & \text{P} \\ & = 16 \text{ atm} & = 8 \text{ atm} & = 4 \text{ atm} \end{array}$$

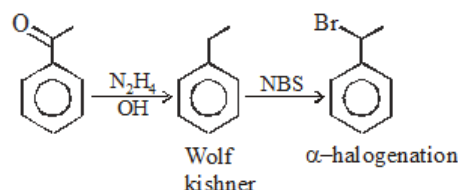
$$K_P = \frac{16^9 \times 8^2 \times 4}{2^4} = K_C (0.08 \times 400)^3$$

$$K_C = 32 \text{ M}^3$$

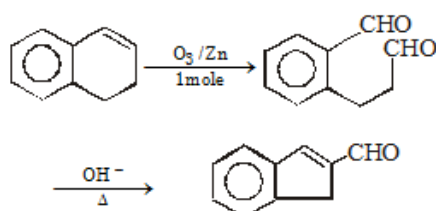
11. **Ans (B)**

HgCl₂ is very weakly ionised in aqueous medium due to more covalent character.

14. **Ans (B)**

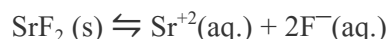
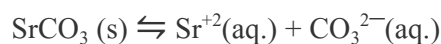


15. **Ans (A)**



SECTION-II

1. **Ans (2.00)**



$$(\text{Sr}^{+2})(\text{CO}_3^{2-}) = 2.5 \times 10^{-10} \dots\dots\dots(1)$$

$$(\text{Sr}^{+2})(\text{F}^-)^2 = 10^{-10} \dots\dots\dots(2)$$

$$(2) / (1) \Rightarrow \frac{(\text{F}^-)^2}{(\text{CO}_3^{2-})} = \frac{10^{-10}}{2.5 \times 10^{-10}}$$

$$[\text{F}^-]^2 = \frac{10^{-10}}{2.5 \times 10^{-10}} \times 10^{-3}$$

$$[\text{F}^-] = 2 \times 10^{-2} \text{ M}$$

2. **Ans (33.33)**



$$t = 0 \quad a \quad -$$

$$t = 10 \quad \frac{a}{2} \quad \frac{a}{4}$$

$$t = 20 \quad \frac{a}{4}$$

$$\frac{3a}{8} \propto 25 \text{ mL}$$

$$\infty \quad - \quad \frac{a}{2} \propto V_\infty$$

$$\frac{3}{8} \times 2 = \frac{25}{V_\infty}$$

3. **Ans (-1.00)**

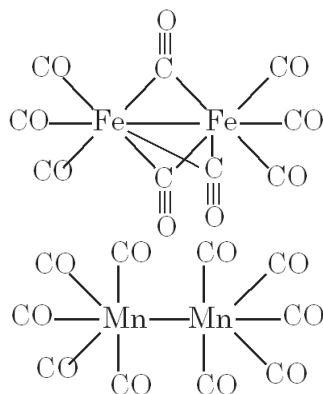
$$(\Delta S)_{\text{uni}} = -10 + 12 = 2 \text{ J/mol-K}$$

$$(\Delta G)_{T,P} = -T(\Delta S)_{\text{uni}}$$

$$= -500 \times 2 = -1000 \text{ J/mol}$$

$$= -1.00 \text{ kJ/mol}$$

5. **Ans (3.00)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

Possible when $AB = BA$

$$\Rightarrow \begin{bmatrix} a-1 & 1-b \\ -a+1 & -1+b \end{bmatrix} = \begin{bmatrix} a-1 & -a+1 \\ 1-b & -1+b \end{bmatrix}$$

$$\Rightarrow a = b$$

2. **Ans (A)**

$$M^2 = I - M,$$

$$M^3 = M(I - M) = M - M^2 = 2M - I$$

$$M^4 = M^3M = 2I - 3M$$

$$\therefore n = 4$$

3. **Ans (B)**

Multiplying the matrices & equating

$$\cos 3\theta = \frac{1}{2} \text{ \& } \sin 3\theta = -\frac{\sqrt{3}}{2}$$

$$\therefore \text{three value of } \theta \in [0, 2\pi]$$

4. **Ans (A)**

Point (1,2) lies on $ay + bx^3 = 10$

$$\Rightarrow 2a + b = 10 \quad \dots(1)$$

$$\text{Also } \frac{dy_1}{dx} \cdot \frac{dy_2}{dx} = -1$$

$$\Rightarrow \left(-\frac{3b}{a}\right) \left(\frac{2}{3}\right) = -1$$

$$\Rightarrow a = 2b \quad \dots(2)$$

5. **Ans (C)**

$$\text{Let } f(x) = x^3 + \alpha x^2 + \beta x + 2$$

$$\therefore f(0) = f(2) \Rightarrow 2 = 8 + 4\alpha + 2\beta + 2$$

$$\Rightarrow 2\alpha + \beta = -4 \quad \dots(1)$$

$$\text{Also, } f'\left(\frac{1}{2}\right) = 3x^2 + 2\alpha + \beta \Big|_{x=\frac{1}{2}}$$

$$\Rightarrow \alpha + \beta = -\frac{3}{4} \quad \dots(2)$$

$$\text{solving (1) \& (2), } \alpha = -\frac{13}{4}, \beta = \frac{5}{2}$$

$$\therefore \alpha + 2\beta = \frac{7}{4}.$$

6. **Ans (C)**

$$f(x) = (a+2)x^3 - 2ax^2 + 4ax - 1$$

$$f'(x) = 3(a+2)x^2 - 4ax + 9a < 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \left. \begin{matrix} a+2 \\ D \leq 0 \end{matrix} \right\} \cap \Rightarrow a \in (-\infty, -3]$$

7. **Ans (A)**

$$y = \left(1 + \frac{1}{x}\right)^{2x} = e^{2x \ln\left(1 + \frac{1}{x}\right)}$$

$$\frac{dy}{dx} = e^{2x \ln\left(1 + \frac{1}{x}\right)} \cdot \left\{ \frac{2x}{1 + (1/x)} \cdot \left(-\frac{1}{x^2}\right) + \ln\left(1 + \frac{1}{x}\right) \cdot 2 \right\}$$

$$\therefore y'(1) = e^{2 \ln 2} \{-1 + 2 \ln 2\} = 4(2 \ln 2 - 1)$$

8. **Ans (C)**

$$\max\left(x, \frac{1}{x}\right) = \begin{cases} x & ; x \in [1, \infty) \\ \frac{1}{x} & ; x \in (0, 1) \\ x & ; x \in [-1, 0) \\ \frac{1}{x} & ; x \in (-\infty, -1) \end{cases}$$

$$\text{and } \min\left(x, \frac{1}{x}\right) = \begin{cases} \frac{1}{x} & ; x \in [1, \infty) \\ x & ; x \in (0, 1) \\ \frac{1}{x} & ; x \in [-1, 0) \\ x & ; x \in (-\infty, -1) \end{cases}$$

$$g(x) = \begin{cases} x^2 & ; x \in [1, \infty) \\ \frac{1}{x^2} & ; x \in (0, 1) \\ x^2 & ; x \in [-1, 0) \\ \frac{1}{x^2} & ; x \in (-\infty, -1) \\ 1 & ; x = 0 \end{cases}$$

9. **Ans (B)**

Best and worst papers occupy cross position in 2 ways

..... × ×

Between these papers there are 3 gaps

Let these gaps are a, b, c

such that

$$a + b + c = 8 \quad \dots(1)$$

$$b \geq 1, a \geq 0, c \geq 0$$

so integral solutions of (i) are 9C_2

and 8 papers are arranged in 8! ways

so total permutations are $2 \cdot {}^9C_2 \cdot 8!$

10. **Ans (D)**

$$[00] * L * D *$$

$$\text{Number of ways} = 3! \times {}^4C_2 = 36$$

11. **Ans (A)**

$$\text{Maximum number of triangles} = {}^{18}C_3 - 3 \cdot {}^6C_3 = 756$$

12. **Ans (D)**

$$\int \frac{x^5 - x}{x^6 + x^2} dx$$

$$= \int \frac{x - \frac{1}{x^3}}{x^2 + \frac{1}{x^2}} dx$$

$$= \frac{1}{2} \ln \left(x^2 + \frac{1}{x^2} \right) + C$$

$$\text{put } x^2 + \frac{1}{x^2} = t$$

13. **Ans (B)**

$$\because \ln(x+1) < x$$

$$\Rightarrow \frac{\ln(x+1)}{x^2} < \frac{1}{x} \Rightarrow I_1 < 1$$

$$\text{Now } x+1 < e^x \Rightarrow \frac{e^x}{(x+1)^2} > \frac{1}{x+1} \Rightarrow I_2 > \ln_2$$

14. **Ans (A)**

$$\lim_{x \rightarrow 0} \frac{\tan^5(3x^2) \cdot 2x}{9 \cdot 12 \cdot x^{11}}$$

$$\left[\frac{\tan(3x^2)}{(3x^2)} \right]^5 \cdot \frac{3^5 \cdot 2}{9 \cdot 12} = \frac{9}{2}$$

15. **Ans (D)**

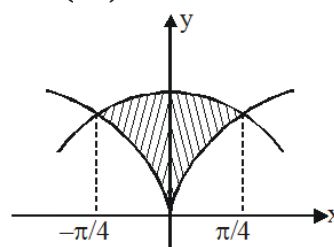
$$\text{Let } I = \int_{-\pi/4}^{\pi/4} \frac{2 \tan^{-1} e^x - \frac{\pi}{2}}{(1+x^2)} dx$$

$$= \int_0^{\pi/4} \frac{2 (\tan^{-1} e^x + \tan^{-1} e^{-x}) - \pi}{(1+x^2)} dx$$

(using property of definite integral)

$$= \int_0^{\pi/4} \frac{2 \cdot \frac{\pi}{2} - \pi}{(1+x^2)} dx = 0$$

16. **Ans (D)**



$$A = 2 \int_0^{\pi/4} (\cos x - \sin x) dx = 2(\sqrt{2} - 1)$$

17. **Ans (C)**

$$\text{Required area} = \left| \frac{16}{3} \cdot \frac{1}{4m} \cdot \frac{1}{4n} \right| = \frac{1}{3}$$

$$\Rightarrow |mn| = 1$$

18. **Ans (A)**

$$y = 2cx + c$$

$$\frac{dy}{dx} = 2c$$

$$\therefore y = x \frac{dy}{dx} + \frac{1}{2} \frac{dy}{dx}$$

$$\therefore 2y = (2x+1) \frac{dy}{dx}$$

19. **Ans (C)**

$$\text{equation can be written as, } \frac{dx}{dy} - y \cdot x = y^3 \cdot x^3$$

$$\text{div. by } x^3 \text{ and put } \frac{1}{x^2} = u, \text{ equation reduces to}$$

linear form.

20. Ans (D)

$$A^2 + AB - 2B^2 = 0$$

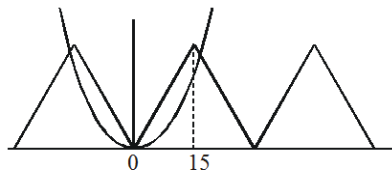
$$(A - B)(A + 2B) = 0$$

$$\Rightarrow A = -2B \Rightarrow AB^{-1} = -2I$$

$$\det(AB^{-1}) = |-2I| = -8$$

SECTION-II

1. Ans (3.00)



Number of solutions are three

2. Ans (47.08)

C-I

$$\underbrace{\quad \quad \quad}_{\text{inc}} \underbrace{\quad \quad \quad}_{\text{dic}} \Rightarrow {}^3C_3 \cdot {}^4C_3 \Rightarrow 4$$

$$\Rightarrow {}^3C_3 \cdot {}^4C_3 \Rightarrow 4$$

C-II

$$\underbrace{\quad \quad \quad}_{\text{inc}} \quad \quad \quad \Rightarrow {}^8C_3 \cdot {}^9C_3$$

$$\Rightarrow {}^8C_3 \cdot {}^9C_3$$

$$\Rightarrow 56.84 \Rightarrow 4704$$

$$\lambda \Rightarrow 4708$$

3. Ans (4.33 or 4.34)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) + 3hx^2 + 3xh^2 - 1}{h}$$

$\therefore$ at $x = y = 0$ in functional rule : $f(0) = 1$

$$\therefore f'(x) = f'(0) + 3x^2 + 0$$

$$\Rightarrow f'(x) = 3x^2 + 1 \Rightarrow f(x) = x^3 + x + C$$

$$\therefore f(0) = 1 \Rightarrow f(x) = x^3 + x + 1$$

$$\frac{f'(2)}{f(1)} = \frac{13}{3} = 4.333$$

4. Ans (7.00)

$$\text{Given } f(x) = \ln \left(\cos x \cdot \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \dots \cos \frac{x}{2^{100}} \right)$$

$$= \ln \left(\frac{\sin 2x}{2^{101} \cdot \sin \frac{x}{2^{100}}} \right)$$

$$\therefore f(x) = \ln \sin 2x - 101 \ln 2 - \ln \sin \frac{x}{2^{100}}$$

$$\Rightarrow f'(x) = 2 \cot 2x - \frac{1}{2^{100}} \cot \frac{x}{2^{100}}$$

$$\Rightarrow f' \left(\frac{\pi}{4} \right) = -\frac{1}{2^{100}} \cot \frac{\pi}{2^{102}}$$

$$\Rightarrow \alpha = 100, \beta = 102$$

$$\therefore (1 + |\alpha - \beta|)^{2015} = (1 + 2)^{2015} = 3^{2015}$$

$$\Rightarrow \text{unit digit} = 7$$

5. Ans (4.00)

$$V = \pi r^2 h$$

$$V = \pi r^2 (5 - r^2)$$

$$V = \pi (5r^2 - r^4)$$

$$\frac{dv}{dr} = \pi (10r - 4r^3) = 0$$

$$r^2 = \frac{5}{2}$$

$$\lambda = \pi \times \frac{5}{2} \left(5 - \frac{5}{2} \right)$$

$$\lambda = \pi \times \frac{25}{4}$$

$$\frac{d^2v}{dr^2} = \pi (10 - 12r^2)$$

$$\frac{d^2v}{dr^2} < 0$$

6. Ans (2.41 or 2.42)

$$\int \frac{\sin^2 x}{\sec^2 x - \cos^2 x} dx$$

$$I = \int \frac{\sin^2 x \cdot \cos^2 x}{1 - \cos^4 x} dx = \int \frac{\cos^2 x}{1 + \cos^2 x} dx$$

$$I = \int \left(1 - \frac{1}{1 + \cos^2 x} \right) dx = \int 1 - \frac{\sec^2 x}{1 + \sec^2 x} dx$$

$$I = x - \int \frac{\sec^2 x}{2 + \tan^2 x} dx$$

$$I = x - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) + C$$

$$\lambda = 1, \mu = \sqrt{2}$$

7. Ans (3.00)

$$\Rightarrow I_1 = \int_0^{\pi} (\pi - x) \sin x e^{\cos 4x} dx \quad (\text{King})$$

$$2I_1 = \pi \int_0^{\pi} \sin x \cdot e^{\cos 4x} dx$$

$$2I_1 = \pi \int_0^{\pi/2} [\sin x \cdot e^{\cos 4x} + \sin x e^{\cos 4x}] dx \quad (\text{Queen})$$

$$I_1 = \pi \int_0^{\pi/2} \sin x \cdot e^{\cos 4x} dx$$

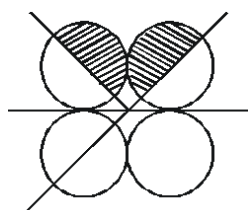
$$= \pi \int_0^{\pi/2} \cos x \cdot e^{\cos 4x} dx \quad (\text{King})$$

$$\left[\frac{I_1}{I_2} \right] = 3$$

8. Ans (9.00)

$$y \geq |x|$$

$$(|x| - 3)^2 + (|y| - 3)^2 \leq 9$$



$\Rightarrow$ full circle.

$\Rightarrow$ area = 9π

k = 9

9. Ans (4.00)

$$y = ax^2 \quad \dots\dots(1)$$

$$\frac{dy}{dx} = 2ax \quad \dots\dots(2)$$

$$\frac{(2)}{(1)} \Rightarrow \frac{dy/dx}{y} = \frac{2}{x} \Rightarrow -\frac{1}{y} \frac{dy}{dx} = \frac{2}{x}$$

$$\Rightarrow 2ydy = -xdx$$

$$y^2 = -\frac{x^2}{2} + C$$

10. Ans (4.00)

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \text{adj}A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Let $|A| = ad - bc = r$, then,

$$|A| + |A|\text{adj}A = 0 = |A| + r.\text{adj}A$$

$$= \begin{vmatrix} a+rd & b(1-r) \\ c(1-r) & d+ar \end{vmatrix} = 0$$

$$\Rightarrow r((r-1)^2 + (a+d)^2) = 0 \Rightarrow r = 1 \text{ and } a$$

$$+ d = 0$$

$$\Rightarrow r((r-1)^2 + (a+d)^2) = 0 \Rightarrow r = 1 \text{ and } a + d = 0$$

$$\text{Now, } |A - r.\text{adj}A| = -(a+d)^2 + 4(ad - bc)$$

$$= 0 + 4r = 4.$$